COURSE SYLLABUS

ENGR 3316: Fluid Dynamics

Fall 2026

RANDOLPH COLLEGE

Instructor	Dr. Huan Phan, Assistant Professor of Robotics and Mechatronics Engineering
Email	hphan206@randolphcollege.edu
Class meetings	**Mon., Tue., Thu., Fri. • 9:30–10:45 a.m. • Martin 507**
Course dates	August 27–October 9, 2026
Office hours	By appointment; weekly times posted on the course website
Course site	huantianh.github.io/courses/fluid-dynamics Announcements, notes, assignments, and schedule updates

Course Description

Fluid dynamics is the study of fluids at rest and in motion and of the forces associated with fluid behavior. This accelerated course introduces the properties of fluids; viscosity and shear stress; pressure variation in static fluids; manometers; hydrostatic forces and buoyancy; conservation of mass; Bernoulli's equation; Eulerian and Lagrangian descriptions of flow; streamlines, pathlines, and streaklines; and Reynolds number and flow regimes. Emphasis is placed on physical interpretation, engineering applications, clear diagrams, correct units, and systematic problem solving.

Expected preparation: Calculus, introductory physics, and engineering mechanics, or permission of the instructor.

Course Learning Outcomes

By the end of the course, a successful student will be able to:

1. explain the defining behavior of a fluid and identify important fluid properties;
2. calculate shear stress, shear force, dynamic viscosity, and kinematic viscosity for Newtonian-fluid applications;
3. determine pressure variation in static fluids and solve gauge, absolute-pressure, and manometer problems;
4. calculate hydrostatic resultant forces, centers of pressure, and buoyant forces on engineering surfaces and bodies;
5. apply conservation of mass and Bernoulli's equation while clearly stating the assumptions and limitations;
6. distinguish Eulerian and Lagrangian descriptions and analyze streamlines, pathlines, and streaklines;
7. select appropriate characteristic velocity and length scales, calculate Reynolds number, and classify flow regimes; and
8. communicate engineering solutions with organized reasoning, diagrams, equations, units, and a physically reasonable final answer.

Course Materials

Required Materials

- Lecture notes, worked examples, readings, and assignments posted on the course website.
- A scientific calculator capable of trigonometric and exponential calculations.
- Paper or a tablet for problem solving during class. Students should bring course notes and a calculator to every meeting.

Recommended Textbook

No textbook purchase is required unless announced by the instructor. The following references are useful supplements:

- Philip M. Gerhart, Andrew L. Gerhart, and John I. Hochstein, *Fundamentals of Fluid Mechanics*, Eighth Edition.

Course Organization and Expectations

The course meets four times per week for 75 minutes. Class meetings combine short lectures, derivations, demonstrations, worked examples, and guided problem solving. Students will regularly work in pairs or small groups to practice course concepts and develop teamwork, communication, and problem-solving skills. Because the course is compressed into seven weeks, students should review notes after each class and begin assignments early.

The course website is the official location for announcements, course materials, assignment instructions, due dates, and schedule changes. Students are responsible for checking it regularly.

Assessment and Grading

Assessment category	Weight
Homework	30%
Midterm Exam	30%
Final Exam	30%
Peer Evaluation	**10%**
Total	**100%**

Unless the instructor announces an adjustment in writing, final course grades will use the following scale:

A	B	C	D	F
90–100%	80–89.9%	70–79.9%	60–69.9%	below 60%

Homework and In-Class Work

Assignments will generally be posted on the course website and will state the exact due date and submission method. Students must show the governing equations, substitutions, units, and enough reasoning

for the work to be followed. An unsupported final answer may receive little or no credit even when numerically correct.

Collaboration in studying and discussing general approaches is encouraged unless an assignment states otherwise. Students will frequently work in pairs or groups during class. Unless group submission is specifically allowed, each student must prepare and submit an independent solution in their own words and identify collaborators when applicable.

Teamwork and Peer Evaluation

Teamwork accounts for 10% of the final grade. Each student will confidentially evaluate their teammates based on preparation, participation, communication, reliability, respect, and contribution. Peer ratings will inform the teamwork grade; the instructor will review the evaluations and determine the final score. Unusually high or low ratings should include a brief explanation.

Examinations and Quizzes

Quizzes may be announced or unannounced and may include conceptual questions or short calculations. The midterm covers the first four major course units. The final examination is comprehensive. Permitted reference materials and calculator rules will be announced before each assessment.

Course Policies

Attendance and Participation

Regular attendance and active participation are expected. Attendance may be recorded. In a seven-week course, one missed meeting represents a substantial amount of material. A student who must miss class should notify the instructor as early as possible, obtain notes and announcements, and remain responsible for missed work. College-approved absences will be handled according to College policy.

Late Work and Make-Up Work

Unless an assignment states otherwise, late work is subject to a deduction of 10 percentage points per calendar day. Students should contact the instructor before the deadline when illness, an emergency, or another serious circumstance will prevent timely submission. Make-up quizzes or examinations are offered only for an approved absence or when arranged with the instructor.

Professional Communication

Students should use their Randolph College email account and include ENGR 3316 in the subject line. The instructor will normally respond within one business day, although messages sent during evenings, weekends, or College holidays may receive a response on the next business day.

Classroom Conduct

Students are expected to contribute to a respectful learning environment. Phones and unrelated electronic devices should be silenced and put away. Laptops and tablets may be used for course-related notes, calculations, and activities.

Randolph College Policies and Resources

Accommodations

Randolph College is committed to providing learning experiences that are accessible for all students and will make reasonable accommodations for students with documented disabilities. A student with a learning difference or a mental-health, medical, or physical disability should contact Larvail Jones, Coordinator of Access Services, at ljones@randolphcollege.edu or 434-947-8132. Students are encouraged to begin this process early. Once an accommodation letter is issued, the student should meet privately with the instructor to discuss its implementation in this course.

Title IX and Sexual Misconduct

As faculty members, instructors are responsible for helping to create a safe learning environment and are mandatory reporters. The instructor will seek to keep information shared by students private to the greatest extent possible, but information regarding sexual misconduct or a crime that may have occurred on Randolph College's campus must be shared with the College's Title IX Coordinator, Jaclyn Beard, at jbeard@randolphcollege.edu or 434-947-8031. The Health Center and Counseling Center staff are confidential resources and are not mandatory reporters; the Counseling Center may be reached at counselingcenter@randolphcollege.edu or 434-947-8130.

Randolph College Honor Code

Integrity is essential in science and engineering. By enrolling in this course, students agree to uphold Randolph College's Honor Code and the policies specific to this course. Students must:

- write submitted answers in their own words and show their own reasoning;
- identify collaborators and cite all sources, software, and outside assistance; and
- be transparent about how submitted work was completed.

Plagiarism, unauthorized assistance, or cheating will be handled under the Randolph College Honor Code. The Student Handbook is available at `https://www.randolphcollege.edu/studenthandbook/`.

Artificial Intelligence

Generative artificial-intelligence tools may be used for brainstorming study questions, improving grammar, or obtaining an additional conceptual explanation. They may not be used to produce a submitted solution, derivation, calculation, diagram, computer output, quiz response, or examination response unless the instructor explicitly authorizes that use for the assignment.

When AI use is permitted, the student remains responsible for checking accuracy, units, assumptions, and sources. Any material contribution from an AI tool must be disclosed by naming the tool and briefly describing how it was used. Undisclosed or unauthorized AI-generated work will be treated as unauthorized assistance under the Honor Code.

Tentative Course Schedule

The schedule may change to match class progress, College announcements, or weather disruptions. The course website will contain the most current version. There is no class on College holidays. The recurring calendar date of Monday, October 12, is a federal holiday and is not a class meeting.

Week	Date	Topic or assessment
1	Thu., Aug. 27	Course introduction; what is a fluid? Fundamental physical laws and fluid properties.
1	Fri., Aug. 28	Viscosity, fluid velocity near a solid surface, and the no-slip condition.
1	Mon., Aug. 31	Shear strain rate; dynamic and kinematic viscosity; Newtonian-fluid shear force.
1	Tue., Sept. 1	Viscosity worked example and guided problem solving.
2	Thu., Sept. 3	Pressure at a point, Pascal's law, and the differential triangular fluid element.
2	Fri., Sept. 4	Hydrostatic equation; pressure at equal elevations; gauge and absolute pressure.
2	Mon., Sept. 7	**No class – Labor Day.**
2	Tue., Sept. 8	Multiple fluids, solid barriers, manometers, and guided practice.
3	Thu., Sept. 10	Hydrostatic force on submerged plane surfaces and vertical walls.
3	Fri., Sept. 11	Center of pressure; gates; inclined and curved surfaces.
3	Mon., Sept. 14	Origin and derivation of buoyant force.
3	Tue., Sept. 15	Floating, sinking, and neutral buoyancy; worked example.
4	Thu., Sept. 17	Bernoulli's equation: motivation, streamline acceleration, and derivation.
4	Fri., Sept. 18	Assumptions, equation forms, energy interpretation, stagnation pressure, and continuity.
4	Mon., Sept. 21	Bernoulli applications, cumulative problem solving, and midterm review.
4	Tue., Sept. 22	**Midterm examination.**
5	Thu., Sept. 24	Eulerian and Lagrangian descriptions of flow; material derivative.
5	Fri., Sept. 25	Streamlines and streamline equations.
5	Mon., Sept. 28	Pathlines, streaklines, and their relationship in steady flow.
5	Tue., Sept. 29	Oscillating-jet worked example and concept questions.
6	Thu., Oct. 1	Reynolds number: physical meaning and selection of characteristic velocity and length.
6	Fri., Oct. 2	Dimensional reasoning, flow regimes, volume flow rate, and mean velocity.
6	Mon., Oct. 5	Reynolds-number applications for pipe flow and external flow.
6	Tue., Oct. 6	Reynolds-number practice, common mistakes, and cumulative course connections.
7	Thu., Oct. 8	Comprehensive review and problem-solving session.
7	Fri., Oct. 9	**Comprehensive final examination.**

The instructor may revise this syllabus when necessary. Any substantive change will be announced in class and posted on the course website.